EXAM SOLUTIONS

1. **These are mostly basic questions directly form slides and course materials, or problems similar to those solved in the class, and require an understanding of basic concepts.**

[3]

Most students referred to the modulation index. About half of them wrote about the effect of noise and very few mentioned carrier to noise ratio. Almost all answered the coherent detection part correctly.

In non-coherent demodulation of AM-modulated signals, when the carrier-to-noise ratio falls below unity, the noise term starts dominating the performance, and the performance of the envelop detector deteriorates very rapidly. This is not the case in coherent demodulation.

ii) [3]

Almost all students answered this question correctly.

It is impossible to demodulate PSK with a non-coherent detector, e.g., envelop detection, since PSK signals have the same frequency and amplitude. However, we can demodulate PSK signals differentially, where phase reference is provided by a delayed version of the signal in the previous interval. This is called differential phase shift keying (DPSK).

iii) [2]

Answers were mostly correct but reasoning was often wrong. A very common wrong answer was "Since the processes are Gaussian they are independent and independence implies uncorrelatedness", and did not refer to "whiteness".

Since it is white, its samples are uncorrelated:

$$E[X(t_1)X(t_2)] = R_N(t_2 - t_1) = 0, \text{ for } t_1 \neq t_2.$$

Since it is also Gaussian, samples are independent.

iv) [2]

Almost all students answered correctly. Very few students wrongly claimed H(Y)=2H(X).

Entropy is only a function of the probability mass function, not the particular values the random variable takes. Therefore, $H(Y) = H(X) = 0.85$ bits.

Mostly correct. **a)** True. A stationary process has time-invariant distribution, and hence, a time-invariant mean. [2]

True/False answers were mostly correct. Some students wrote the Hilbert transform expression, but could not interpret its effect on a WSS process.

ii) True. WSS is preserved going through a linear time invariant (LTI) system. [2]

Mostly correct. iii) False. Being Gaussian is about the distribution, it does not specify the power spectral density. [2]

iv) True. We have $E[X(t)Y(t)] = E[X(t)]E[Y(t)] = \text{constant}$, and

About a third of the students, after showing that E(XY)=E(X)E(Y) is time invariant, directly claimed "True" without checking R_XY.

$$\begin{aligned} R_{XY}(t_1,t_2) &= E[X(t_1)Y(t_1)X(t_2)Y(t_2)] \\ &= E[X(t_1)X(t_2)]E[Y(t_1)Y(t_2)] \\ &= R_X(t_2 - t_1)R_Y(t_2 - t_1). \end{aligned}$$

[3]

b) i) [2]

Mostly correct.

With $m_p = 3$, the maximum quantization error is given by $\Delta/2 = 2m_p/2L = 3/256 = 0.0117$.

ii) [3]

Mostly correct. Some minor mistakes in signal power calculation and using Δ^3 instead of Δ^2.

The SNR at the quantizer output is given by $SNR = \frac{3P}{m_p^2} 2^{2n}$, where P is the average power of the signal and n is the number of bits per sample, i.e., $n = \log_2 L$. We have $P = A_m^2/2 = 9/2$, where $A_m = 3$, and $n = 8$ bits. Therefore, we obtain $SNR = 3 \cdot 2^{15} \sim 49.9$ dB.

iii) [2]

Students mostly found the Nyquist frequency correctly. A common mistake was using Hz as the unit instead of kbps.

Nyquist frequency is 8 kHz. The sampling frequency is 10 kHz, leading to the PCM bandwidth of $8 * 10 = 80$ kbits per second.

iv) [3]

If the sampling frequency is 14 kHz, we should have $14n = 80$; hence, we can have at most 5 bits per sample.

This would result in $SNR = 3 \cdot 2^9 \sim 31.9$ dB. Note that a reduction of 3 bits per samle results in 3x6=18 dB reduction in the SNR.

c) We have

Student mostly seem to understand WSS processes and how to find the second moment. Common mistakes were about taking the inverse transform of the PSD due to wrong shifts.

$$\begin{aligned} S_X(f) &= \frac{\pi/2}{\left(\frac{\pi}{2}\right)^2 + \left(2\pi\left(f - \frac{1}{4}\right)\right)^2} + \frac{\pi/2}{\left(\frac{\pi}{2}\right)^2 + \left(2\pi\left(f + \frac{1}{4}\right)\right)^2} \\ &= \left[\frac{2(\pi/2)}{\left(\frac{\pi}{2}\right)^2 + (2\pi f)^2}\right] * \frac{1}{2}\left[\delta(f - \frac{1}{2}) + \delta(f + \frac{1}{4})\right] \end{aligned}$$

The autocorrelation of the given WSS random process can be found by taking the inverse Fourier transform of the given PSD.

$$R_X(\tau) = e^{-\frac{\pi}{2}|\tau|} \cdot \cos\left(\frac{\pi}{2}\tau\right).$$

For a real WSS process, we have

$$\begin{aligned} E\left[\left(X(t+\tau) - X(t)\right)^2\right] &= E[X^2(t+\tau) - 2X(t+\tau)X(t) + X^2(t)] \\ &= 2[R_X(0) - R_X(\tau)]. \end{aligned}$$

i) [5]

There were mostly calculation errors due to wrong multiplication. However, most students seem to understand the basics of WSS processes.

For $\tau = 1/\pi$, we have

$$2[R_X(0) - R_X(\tau)] = 2[1 - e^{-1/2}\cos(1/2)].$$

ii) [6]

$$\begin{aligned}
&\mathrm{E}\left[(X(t+1)+X(t+3)-X(t+5)-X(t))^2\right] \\
&\quad = \mathrm{E}[X^2(t+1)+X^2(t+3)+X^2(t+5)+X^2(t) \\
&\quad +2X(t+1)X(t+3)-2X(t+1)X(t+5)-2X(t+1)X(t) \\
&\quad -2X(t+3)X(t+5)-2X(t+3)X(t)+2X(t+5)X(t) \\
&= 4R_X(0)-2R_X(4)-2R_X(1)-2R_X(3)-2R_X(5).
\end{aligned}$$

We note that $R_X(1)=R_X(3)=R_X(5)=0$ since $\cos(\tau\pi/2)=0$ when τ is an odd integer. Therefore, the second moment is given by

$$4R_X(0)+2R_X(4)=4-2e^{-2\pi}.$$

2. **The first part is similar to problems solved in the class, and expects a good understanding of the fundamental conceipts relating to random processes. The second one is a different problem that will require some deeper thinking and manipulation of the concepts learned in the lectures.**

a) [5]

We have

Many students did not mention the first condition of wide sense stationarity; that is, the mean does not depend on time.

$$\begin{aligned}
E[X(t)] &= E[N_1\cos(2\pi f_c t)-N_2\sin(2\pi f_c t)] \\
&= E[N_1]\cos(2\pi f_c t)-E[N_2]\sin(2\pi f_c t) \\
&= 0
\end{aligned}$$

It seems there is a common misunderstanding of randomness. The frequent mistake was taking the expectation also for sine and cosine, which are deterministic functions of time.

Few students could not see E(N1N2)=E(N1)E(N2) due to their independence.
Another common mistake was using the wrong formula for the identity sin(a)*sin(b) even though though it was provided.
A less frequent mistake was the definition of autocorrelation function. Some students used E[x(t)x(t)], which is actually Rx(0).

$$\begin{aligned}
R_X(t_1,t_2) &= E[X(t_1)X(t_2)] \\
&= E\left[(N_1\cos(2\pi f_c t_1)-N_2\sin(2\pi f_c t_1))(N_1\cos(2\pi f_c t_2)-N_2\sin(2\pi f_c t_2))\right] \\
&= E[N_1^2]\cos(2\pi f_c t_1)\cos(2\pi f_c t_2)+E[N_2^2]\sin(2\pi f_c t_1)\sin(2\pi f_c t_2) \\
&\quad -E[N_1N_2](\cos(2\pi f_c t_1)\sin(2\pi f_c t_2)+\sin(2\pi f_c t_1)\cos(2\pi f_c t_2)) \\
&= \sigma^2[\cos(2\pi f_c t_1)\cos(2\pi f_c t_2)+\sin(2\pi f_c t_1)\sin(2\pi f_c t_2)] \\
&= \sigma^2\cos(2\pi f_c(t_1-t_2)),
\end{aligned}$$

where we have used the linearity of the expectation operation, and the independence of N_1 and N_2, i.e., $E[N_1N_2]=E[N_1]E[N_2]=0$. We can write the autocorrelation function as follows:

$$R_X(\tau)=\sigma^2\cos(2\pi f_c\tau).$$

ii) [2]

Since the mean is independent of time, and the autocorrelation function depends only on the time difference, $X(t)$ is WSS.

iii) [4]

We have

Most students knew the relation between the autocorrelation function and the power spectral density.

$$\begin{aligned} S_X(f) &= \mathbb{F}(R_X(\tau)) \\ &= \mathbb{F}\left(\sigma^2 \cos(2\pi f_c \tau)\right) \\ &= \frac{\sigma^2}{2}[\delta(f-f_c)+\delta(f+f_c)]. \end{aligned}$$

iv) [4]

More than half of the students could not write Sy(f)= |H(f)|^2 Sx(f)

Among those who got this correctly, many could not conclude $(-\text{j sgn(f)})^2=1$

We can consider $Y(t)$ as the output of an LTI filter with frequency response $H(f) = -j \cdot \text{sgn}(f)$. Then the PSD of $Y(t)$ is found as:

$$\begin{aligned} S_Y(f) &= |H(f)|^2 S_X(f) \\ &= |-j \cdot \text{sgn}(f)|^2 S_X(f) \\ &= \frac{\sigma^2}{2}[\delta(f-f_c)+\delta(f+f_c)]. \end{aligned}$$

b) i) [3]

There are very few students getting the noise variance correct.

Most students wrote the error probability as in the symmetric case.

In FSK, we have $y_1(t) = A + [n_1(t) - n_0(t)]$ and $y_0(t) = -A + [n_1(t) - n_0(t)]$. In this case the noise variance is given by $\sigma^2 = N_0/T_b + 2N_0/T_b = 3N_0/T_b$. Error probability is $Q\left(\frac{A}{\sigma}\right) = Q\left(\sqrt{\frac{2E_b}{3N_0}}\right)$.

ii) [4]

Error probability is $Q\left(\frac{3A}{2\sigma}\right)$.

Most students got Eb correctly, but some of them assumed the same error probability expression as in the pervious part.

Average energy per bit is $E_b = \frac{1}{2}\frac{A^2T_b}{2} + \frac{1}{2}\frac{4A^2T_b}{2} = \frac{5A^2T_b}{4}$.

The noise variance as before $\sigma^2 = 3N_0/T_b$. Therefore, the error probability can be obtained as $Q\left(\sqrt{\frac{3E_b}{5N_0}}\right)$.

This error probability is higher than the one in part i), because symmetric signalling is always more efficient in terms of average energy.

iii) [4]

Another common mistake was about the Q function. There is a common misconception that Q(x) is an increasing function of x.

More students got part iii) compared to parts i) and ii). The common mistake here was not realizing that the channel with f0 has lower noise variance.

We should transmit pulses of magnitudes A and $-A$ over frequency f_0, where the noise variance is lower. This would result in standard PSK with an error probability of $Q(A/\sigma)$, where $E_b = \frac{A^2T_b}{2}$ and $\sigma^2 = N_0/T_b$. Equivalently, the error probability is given by $Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$. This error probability is lower than both of the schemes in parts (i) and (ii).

iv) [4]

Mostly correct, but some students misinterpreted the question and proposed to compress the source.

There are many ways to do this. We can transmit a pulse over frequency f_0 and another pulse over frequency f_1. Or, we can also use FSK as in part (i), but can have pulses of 4 different amplitudes, e.g., $(-3A, -A, +A, +3A)$.

3. **The first part is an unseen problem and requires the students to understand the impact of basic components of a demodulator, and connect it to the concepts learned in the class. Part b) has both a conceptual part and a more strightforwrd evaluation part. The last part can be answered based on basic probbaility knowledge, or directly using the answer to the conceptual question.**

a) [3]

We have

$$x_1(t) = A\cos(2\pi ft)\cos(2\pi f_1 t)$$
$$= \frac{A}{2}\cos(2\pi(f-f_1)t) + \frac{A}{2}\cos(2\pi(f+f_1)t)$$

and

Mostly answered correctly. Some students made a mistake in the cos() * sin() expression.

$$x_2(t) = A\cos(2\pi ft)\sin(2\pi f_1 t)$$
$$= -\frac{A}{2}\sin(2\pi(f-f_1)t) + \frac{A}{2}\sin(2\pi(f+f_1)t)$$

ii) [4]

We have

$$y_1(t) = \frac{A}{2}\cos(2\pi(f-f_1)t)$$

Answers were mostly correct.
Very few students took the high frequency component instead of the low frequency one.

and

$$y_2(t) = -\frac{A}{2}\sin(2\pi(f-f_1)t)$$

since $f + f_1 = f + (f_L + f_H)/2$ is greater than $W = (f_H + 2f_L)/2$ for any $f \in [f_L, f_H]$.

iii) [3]

Here, the common problem was that students either removed all the frequencies and ignored more terms than they should, or kept everything which could not be reduced to a single term.

$$u(t) = \frac{A}{2}\cos(2\pi(f-f_1)t)\cos(2\pi f_2 t) - \frac{A}{2}\sin(2\pi(f-f_1)t)\sin(2\pi f_2 t)$$
$$= \frac{A}{2}\cos(2\pi(f_2 + f - f_1)t)$$

iv) [5]

Very common wrong answers were AM modulation and DSB. Mostly claimed the carrier frequency as f2. This was due to mistakenly removing extra terms in part iii.

This is upper single sideband modulation (SSB) with a carrier frequency of $f_2 - f_1$.

b) i) [4]

The joint entropy of two random variables is always smaller than or equal to the sum of their individual entropies. $H(X,Y)$ and $H(X) + H(Y)$ are equal if and only if X and Y are independent.

Mostly correct, but some students wrongly claimed that H(X,Y)< H(X) + H(Y) when X and Y are independent, and H(X ,Y)>H (X)+H(Y) if not.

ii) [4]

The marginal distribution of X is given by $P(X=1)=P(X=2)=5/16$ and $P(X=3)=6/16$. Therefore, we have

$$\begin{aligned} H(X) =& \frac{5}{16}\log\frac{16}{5}+\frac{6}{16}\log\frac{16}{6} \\ =& 4-\frac{5}{8}\log 5-\frac{3}{8}(1+\log 3) \\ =& 1.579 \end{aligned}$$

The first two parts were answered correctly by most of the students.

Y has the exact same marginal distribution, and hence, the same entropy.

iii) [4]

The joint entropy can be calculated from the joint distribution as follows:

$$\begin{aligned} H(X,Y) =& 3\times\frac{1}{16}\log\frac{16}{1}+\frac{2}{16}\log\frac{16}{2}+\frac{3}{16}\log\frac{16}{3}+2\times\frac{4}{16}\log\frac{16}{4} \\ =& 4-\frac{1}{8}\log 2-\frac{3}{16}\log 3-\frac{8}{16}\log 4 \\ =& 2.578 \end{aligned}$$

iv) [3]

Students answering part i correctly answered this one as well. Few students used the joint distribution to justify their answer.

If they were independent, the joint entropy would be equal to the sum of individual entropies, which is not the case. Moreover, the fact that they are not independent is also evident from their joint distribution, which is not equal to the multiplication of their marginals.